

**SIMATS ENGINEERING  
ASSESSMENT TOOL 3**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1710

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**COURSE OUTCOME COVERED**

*CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 20%*

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**QUESTIONS:** 5

**Time Duration:** 1 Hour  
**Total Marks:** 30

Annexure A

**Resolution Algorithm Implementation**

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**Code of Conduct:**

I Kanish Kumar.S (192511415) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

S. Kanish Kumar

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## CD3 - AT3

### RESOLUTION ALGORITHM IMPLEMENTATION

#### ① MEDICAL ATTENTION ALERT:

1. Propositional Logic,

Let,

$F$  : Patient has a fever

$M$  : Patient needs medical Attention

Knowledge Base:

\* If patient has fever, then patient needs medical attention

$$F \rightarrow M$$

\* Ravi has a fever

$F$

Goal: Prove  $M$

2. Convert to CNF,

$F \rightarrow M$  becomes

$$\neg F \vee M$$

Fact:

$F$

Negate the goal:  $\neg M$

So, the clauses are,

\*  $C1: \neg F \vee M$

\*  $C2: F$

\*  $C3: \neg M$

3. Resolution steps,

Resolve  $C1$  and  $C2$ ,

$$(\neg F \vee M) \vee F$$

Using  $F$  and  $\neg F$ :

$M$

Now resolve.

M with  $\neg M$

Therefore,

$\square$  (Empty clause)

4. Conclusion,

Since the empty is derived, Ravi needs medical attention.

②

LIBRARY LATE FINE:

1. Propositional Logic,

Let,

L = Book was returned late

F = Fine is charged

Knowledge Base

$L \rightarrow F$   
L

Goal: Prove F

2. CNF,

$L \rightarrow F$

becomes.

$\neg L \vee F$

Facts:

L

Negated goal:  $\neg F$

Clauses:

\* C1:  $\neg L \vee F$

\* C2: L

\* C3:  $\neg F$

3. Resolution steps,

Resolve (1) and (2):

$(L \rightarrow L \vee F)$  and  $L$

Therefore,

$F$

Resolve  $F$  with the negated goal

$F$  and  $\neg F$

Therefore,

□ (Empty clause)

4. Conclusion,

Hence, a fine is charged.

③ FLIGHT DELAY NOTIFICATION:

1. Propositional Logic,

Let,

$D$  = Flight is delayed.

$N$  = Passengers are noticed.

Knowledge Base:

$D \rightarrow N$

$N$

Goal: Prove  $N$

2. CNF,

$D \rightarrow N$

becomes,

$\neg D \vee N$

Fact:

$D$

Negated goal :

$\neg N$

clauses :

\*  $C_1 : \neg D \vee N$

\*  $C_2 : D$

\*  $C_3 : \neg N$

3. Resolution steps,

Resolve  $C_1$  and  $C_2$  :

$(\neg D \vee N)$  and  $D$

Therefore,

$N$

Resolve  $N$  with  $\neg N$  :

$N$  and  $\neg N$

Therefore,

□ (Empty clause)

4. Conclusion,

Hence, passengers are notified.

#### ④ EMPLOYEE CERTIFICATION :

1. Propositional Logic,

Let,

$T$  = Kevin completed the training.

$C$  = Kevin receives certification.

Knowledge Base :

$T \rightarrow C$

$T$

Goal : Prove  $C$



CNF,

$T \rightarrow C$

becomes:

$\neg T \vee C$

Fact:

$T$

Negated goal:  $\neg C$

clauses:

\*  $C1: \neg T \vee C$

\*  $C2: T$

\*  $C3: \neg C$

3. Resolution steps,

Resolve  $C1$  and  $C2$ :

$(\neg T \vee C)$  and  $T$

Therefore,

$C$

Resolve  $C$  with  $\neg C$ :

$C$  and  $\neg C$

Therefore,

□ (Empty clause)

4. Conclusion,

Hence, Kevin receives a Certification.

## ⑤ EMPLOYEE BENEFITS ELIGIBILITY:

1. Propositional Logic,

Let,

\*  $P$  = Employee finished probation.

\*  $S$  = Employee becomes permanent staff.

\*  $B$  = Employee gets full benefits.

## Knowledge Base

1.  $P \rightarrow S$

2.  $S \rightarrow B$

3.  $P$

Goal : Prove  $B$

2. Convert Implications into CNF,

$$P \rightarrow S$$

becomes :

$$\neg P \vee S$$

$$S \rightarrow B$$

becomes :

$$\neg S \vee B$$

Fact :  $P$

Negated goal :  $\neg B$

clauses :

\*  $C_1 : \neg P \vee S$

\*  $C_2 : \neg S \vee B$

\*  $C_3 : P$

\*  $C_4 : \neg B$

3. Resolution Steps,

Step 1 : Resolve  $C_1$  and  $C_3$ ,

$$(\neg P \vee S) \text{ and } P$$

Therefore,  $S$ .

Step 2 : Resolve  $C_2$  with  $S$  :

$$(\neg S \vee B) \text{ and } S$$

Therefore,  $B$ .

Step 3 : Resolve  $B$  with the negated goal  $\neg B$  :

$$B \text{ and } \neg B$$

Therefore,  $\square$  (Empty clause)

4. Final Conclusion : Since, the Empty clause ( $\square$ ) is derived.

Therefore, the employee gets full benefits.

### RUBRICS FOR CALCULATION

| Criteria                | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                     | Level 1 – Beginning                                  | Marks (100) |
|-------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|------------------------------------------|------------------------------------------------------|-------------|
| Understanding of Case   | 25        | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding; problem not identified correctly | 22          |
| Application of Concepts | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts          | Fails to apply relevant concepts                     | 24          |
| Analysis                | 20        | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth        | Weak or no analysis; lacks reasoning                 | 19          |
| Solution Design         | 20        | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification  | Incorrect or no solution provided                    | 18          |
| Clarity                 | 10        | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure            | Poor clarity; difficult to understand                | 8           |

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